variability or dispersion of the samples taken from the given process. The control limits of the $\bar{X}$ and R charts shows the presence or absence of assignable causes in the production process. Both $\bar{X}$ and R charts are usually required for decision making to accept or reject the process.

The procedure for constructing $\bar{X}$ and R charts are outlined below.

Procedure for $\bar{X}$

(i) Let X_1, X_2, X_3, etc. be the samples selected, each containing 'n' observations (usually n = 4, 5 or 6)

(ii) Calculate mean for each samples $\bar{X}_1, \bar{X}_2, \bar{X}_3 \ldots$ by using

$$\bar{X}_i = \frac{\sum X_i}{n}, i = 1, 2, 3, 4, \ldots$$

where $\sum X_i$ = total of 'n' values included in the sample X_i.

(iii) Find the mean $\left(\bar{\bar{X}}\right)$ of the sample means.

$$\bar{\bar{X}} = \frac{\sum \bar{X}}{number\ of\ sample\ means}$$

where $\sum \bar{X}$ = total of all the sample means.

Procedure for R -Charts.

Calculate $R = x_{max} - x_{min}$

Let $R_1, R_2, R_3 \ldots$ be the ranges of the 'n' samples. The average range is given by

$$\bar{R} = \frac{\sum R}{n}$$

The calculation of control limits for $\bar{X}$ chart in two different cases is

Case (i) when $\bar{X}$ and SD are given	Case (ii) when $\bar{X}$ and SD are not given
$UCL = \bar{\bar{X}} + 3\frac{\sigma}{\sqrt{n}}$	$UCL = \bar{\bar{X}} + A_2\bar{R}$
$CL = \bar{\bar{X}}$	$CL = \bar{\bar{X}}$
$LCL = \bar{\bar{X}} - 3\frac{\sigma}{\sqrt{n}}$	$LCL = \bar{\bar{X}} - A_2\bar{R}$

Table 9.18

The calculation of control limits for R chart in two different cases are

Case (i) when SD is given	Case (ii) when SD is not given
$UCL = \bar{R} + 3\sigma_R$	$UCL = D_4\bar{R}$
$CL = \bar{R}$	$CL = \bar{R}$
$LCL = \bar{R} - 3\sigma_R$	$LCL = D_3\bar{R}$

Table 9.19

The values of A_2, D_3 and D_4 are given in the table.

Example 9.19

A machine drills hole in a pipe with a mean diameter of 0.532 cm and a standard deviation of 0.002 cm. Calculate the control limits for mean of samples 5.

Solution:

Given $\bar{\bar{X}} = 0.532$, $\sigma = 0.002$, $n = 5$

The control limits for $\bar{X}$ chart is

$$UCL = \bar{\bar{X}} + 3\frac{\sigma}{\sqrt{n}} = 0.532 + 3\frac{0.002}{\sqrt{5}} = 0.5346$$

$$CL = \bar{\bar{X}} = 0.532$$

$$UCL = \bar{\bar{X}} - 3\frac{\sigma}{\sqrt{n}} = 0.532 - 3\frac{0.002}{\sqrt{5}} = 0.5293$$

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